EDEXCEL INTERNATIONAL GCSE MATH (4MA1) • UNIT 4.01

Coordinate Geometry & Linear Programming

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

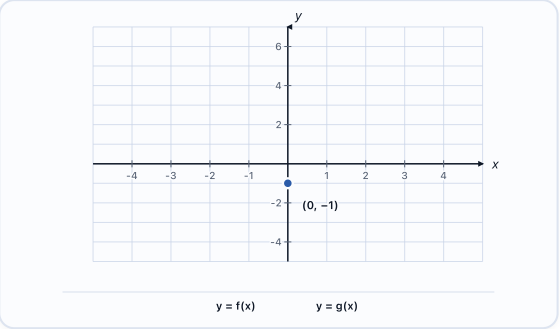
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 D	Q2 A	Q3 B	Q4 C	Q5 D
Q6 A	Q7 B	Q8 C	Q9 C	Q10 B
Q11 D	Q12 B	Q13 B	Q14 B	Q15 C
Q16 C	Q17 D	Q18 A	Q19 C	Q20 B
Q21 D	Q22 C	Q23 D	Q24 D	Q25 B

Coordinate Geometry & Linear Programming Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 12x + 50$ in the domain of Coordinate Geometry & Linear Programming. By completing the square $f_1(x) = (x-h)^2 + k$, find the sum $k + f_1(8)$.



A 38

B 18

C 30

D 32 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Coordinate Geometry & Linear Programming):

1. Complete the square: $x^2 - 12x + 50 = (x-6)^2 - 36 + 50 = (x-6)^2 + 14$, giving minimum value $k = 14$. **[M1]**
2. Evaluate at $x = 8$: $f_1(8) = (2)^2 + 14 = 18$. **[M1]**
3. Sum the two quantities: $k + f_1(8) = 14 + 18 = 32$. **[A1]**

- **Final Verified Answer:** 32.

2 MCQ

A mathematical model in Coordinate Geometry & Linear Programming is defined by the cubic function $y_2(x) = 2x^3 - 6x^2 + 15x - 11$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

- **A** 33 ✓
- **B** 69
- **C** 18
- **D** 21

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Coordinate Geometry & Linear Programming):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 6x^2 - 12x + 15$. **[M1]**
2. Substitute $x = 3$: $\left.\frac{dy}{dx}\right|_{x=3} = 6(9) - 12(3) + 15 = 33$. **[M1][A1]**

- **Final Verified Answer:** 33.

3 MCQ

In Coordinate Geometry & Linear Programming, two displacement vectors are given by $\mathbf{u}_3 = \langle 6, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

- **A** -2
- **B** 98 ✓
- **C** 100
- **D** 210

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Coordinate Geometry & Linear Programming):

1. Compute the inner product: $\mathbf{u}_3 \cdot \mathbf{v}_3 = (6)(9) + (-8)(7) = 54 + (-56) = -2$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_3\|^2 = 6^2 + (-8)^2 = 100$. **[M1]**
3. Combine: $-2 + 100 = 98$. **[A1]**

- **Final Verified Answer:** 98.

4 MCQ

The linear growth profile shown in the diagram for Coordinate Geometry & Linear Programming follows an arithmetic progression with first term $a_1 = 7$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

- **A** 91
- **B** 825
- **C** 735 ✓
- **D** 1365

WORKED SOLUTION & VERIFICATION

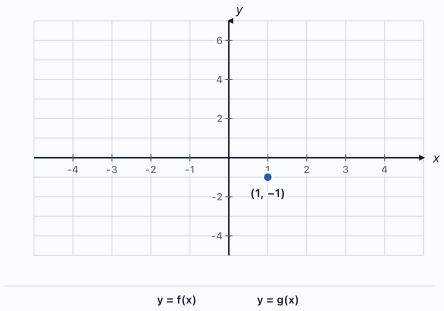
Mark Scheme & Examiner Solution (Coordinate Geometry & Linear Programming):

1. Use the sum formula for an arithmetic series: $S_n = \frac{n}{2}[2a_1 + (n-1)d]$. **[M1]**
2. Substitute $n = 15, a_1 = 7, d = 6$: $S_{15} = \frac{15}{2}[14 + 14 \times 6] = \frac{15}{2} \times 98 = 735$. **[M1][A1]**

- **Final Verified Answer:** 735.

5 MCQ

In Coordinate Geometry & Linear Programming, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 24x)\,\mathrm{d}x$.



A 120

B 120 + 1

C 49

D 96 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Coordinate Geometry & Linear Programming):

1. Integrate term by term: $\int (18x^2 + 24x)\,\mathrm{d}x = [6x^3 + 12x^2]_0^2$. **[M1]**
2. Substitute the upper and lower limits: $(6(2)^3 + 12(2)^2) - 0 = 48 + 48 = 96$. **[M1][A1]**

- **Final Verified Answer:** 96.

6 MCQ

In Coordinate Geometry & Linear Programming, consider the quadratic function $f_6(x) = x^2 - 14x + 135$. Express $f_6(x)$ in the completed-square form $(x - h)^2 + k$ and compute the exact value of $k + f_6(14)$.

A 221 ✓

B 228

C 135

D 214

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Coordinate Geometry & Linear Programming):

1. Complete the square: $x^2 - 14x + 135 = (x - 7)^2 - 49 + 135 = (x - 7)^2 + 86$, giving minimum value $k = 86$. **[M1]**
2. Evaluate at $x = 14$: $f_6(14) = (7)^2 + 86 = 135$. **[M1]**
3. Sum the two quantities: $k + f_6(14) = 86 + 135 = 221$. **[A1]**

- **Final Verified Answer:** 221.

7 MCQ

The graph illustrates the cubic model $y_7(x) = 2x^3 - 7x^2 + 33x - 18$ studied in Coordinate Geometry & Linear Programming. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

A 417

B 305 ✓

C 272

D 67

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Coordinate Geometry & Linear Programming):

1. Differentiate term by term using the power rule: $\frac{\mathrm{d}y}{\mathrm{d}x} = 6x^2 - 14x + 33$. **[M1]**
2. Substitute $x = 8$: $\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=8} = 6(64) - 14(8) + 33 = 305$. **[M1][A1]**

- **Final Verified Answer:** 305.

8 MCQ

In Coordinate Geometry & Linear Programming, two displacement vectors are given by $\mathbf{u}_8 = \langle 7, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 21

B 98

C 119 ✓

D 189

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Coordinate Geometry & Linear Programming):

1. Compute the inner product: $\mathbf{u}_8 \cdot \mathbf{v}_8 = (7)(8) + (-7)(5) = 56 + (-35) = 21$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_8\|^2 = 7^2 + (-7)^2 = 98$. **[M1]**
3. Combine: $21 + 98 = 119$. **[A1]**

- **Final Verified Answer:** 119.

9 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(0, 2)$ and $B(2, 8)$ (Segment #1).

A $y = -\frac{1}{3}x + 9$

B $y = \frac{1}{3}x - 2$

C $y = -\frac{1}{3}x + \frac{16}{3}$ ✓

D $y = 3x + 5$

WORKED SOLUTION & VERIFICATION

Midpoint $M = (1, 5)$ and gradient $m_{AB} = \frac{6}{2} = 3$. The perpendicular gradient is $m_\perp = -\frac{1}{3}$, giving $y - 5 = -\frac{1}{3}(x - 1)$, which simplifies to $y = -\frac{1}{3}x + \frac{16}{3}$.

10 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(2, 4)$ and $B(6, 12)$ (Segment #2).

A $y = \frac{1}{2}x - 5$

B $y = -\frac{1}{2}x + 10$ ✓

C $y = -\frac{1}{2}x + 15$

D $y = 2x + 8$

WORKED SOLUTION & VERIFICATION

Midpoint $M = (4, 8)$ and gradient $m_{AB} = \frac{8}{4} = 2$. The perpendicular gradient is $m_\perp = -\frac{1}{2}$, giving $y - 8 = -\frac{1}{2}(x - 4)$, which simplifies to $y = -\frac{1}{2}x + 10$.

11 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(4, 6)$ and $B(10, 8)$ (Segment #3).

A $y = 3x - 8$

B $y = -3x + 17$

C $y = \frac{1}{3}x + 7$

D $y = -3x + 28$ ✓

WORKED SOLUTION & VERIFICATION

Midpoint $M = (7, 7)$ and gradient $m_{AB} = \frac{2}{6} = \frac{1}{3}$. The perpendicular gradient is $m_\perp = -3$, giving $y - 7 = -3(x - 7)$, which simplifies to $y = -3x + 28$.

12 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(6, 8)$ and $B(14, 12)$ (Segment #4).

- A $y = -2x + 23$
- B $y = -2x + 30$ ✓
- C $y = \frac{1}{2}x + 10$
- D $y = 2x - 11$

WORKED SOLUTION & VERIFICATION

Midpoint $M = (10, 10)$ and gradient $m_{AB} = \frac{4}{8} = \frac{1}{2}$. The perpendicular gradient is $m_{\perp} = -2$, giving $y - 10 = -2(x - 10)$, which simplifies to $y = -2x + 30$.

13 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(8, 10)$ and $B(10, 16)$ (Segment #5).

- A $y = \frac{1}{3}x - 10$
- B $y = -\frac{1}{3}x + 16$ ✓
- C $y = 3x + 13$
- D $y = -\frac{1}{3}x + 25$

WORKED SOLUTION & VERIFICATION

Midpoint $M = (9, 13)$ and gradient $m_{AB} = \frac{6}{2} = 3$. The perpendicular gradient is $m_{\perp} = -\frac{1}{3}$, giving $y - 13 = -\frac{1}{3}(x - 9)$, which simplifies to $y = -\frac{1}{3}x + 16$.

14 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(10, 0)$ and $B(14, 8)$ (Segment #6).

- A $y = -\frac{1}{2}x + 19$
- B $y = -\frac{1}{2}x + 10$ ✓
- C $y = 2x + 4$
- D $y = \frac{1}{2}x - 13$

WORKED SOLUTION & VERIFICATION

Midpoint $M = (12, 4)$ and gradient $m_{AB} = \frac{8}{4} = 2$. The perpendicular gradient is $m_{\perp} = -\frac{1}{2}$, giving $y - 4 = -\frac{1}{2}(x - 12)$, which simplifies to $y = -\frac{1}{2}x + 10$.

15 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(12, 2)$ and $B(18, 4)$ (Segment #7).

- A $y = -3x + 21$
- B $y = 3x - 16$
- C $y = -3x + 48$ ✓
- D $y = \frac{1}{3}x + 3$

WORKED SOLUTION & VERIFICATION

Midpoint $M = (15, 3)$ and gradient $m_{AB} = \frac{2}{6} = \frac{1}{3}$. The perpendicular gradient is $m_{\perp} = -3$, giving $y - 3 = -3(x - 15)$, which simplifies to $y = -3x + 48$.

16 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(0, 4)$ and $B(8, 8)$ (Segment #8).

A $y = 2x - 5$

B $y = \frac{1}{2}x + 6$

C $y = -2x + 14$ ✓

D $y = -2x + 13$

WORKED SOLUTION & VERIFICATION

Midpoint $M = (4, 6)$ and gradient $m_{AB} = \frac{4}{8} = \frac{1}{2}$. The perpendicular gradient is $m_{\perp} = -2$, giving $y - 6 = -2(x - 4)$, which simplifies to $y = -2x + 14$.

17 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(2, 6)$ and $B(4, 12)$ (Segment #9).

A $y = 3x + 9$

B $y = \frac{1}{3}x - 4$

C $y = -\frac{1}{3}x + 15$

D $y = -\frac{1}{3}x + 10$ ✓

WORKED SOLUTION & VERIFICATION

Midpoint $M = (3, 9)$ and gradient $m_{AB} = \frac{6}{2} = 3$. The perpendicular gradient is $m_{\perp} = -\frac{1}{3}$, giving $y - 9 = -\frac{1}{3}(x - 3)$, which simplifies to $y = -\frac{1}{3}x + 10$.

18 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(4, 8)$ and $B(8, 16)$ (Segment #10).

A $y = -\frac{1}{2}x + 15$ ✓

B $y = \frac{1}{2}x - 7$

C $y = -\frac{1}{2}x + 21$

D $y = 2x + 12$

WORKED SOLUTION & VERIFICATION

Midpoint $M = (6, 12)$ and gradient $m_{AB} = \frac{8}{4} = 2$. The perpendicular gradient is $m_{\perp} = -\frac{1}{2}$, giving $y - 12 = -\frac{1}{2}(x - 6)$, which simplifies to $y = -\frac{1}{2}x + 15$.

19 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(6, 10)$ and $B(12, 12)$ (Segment #11).

A $y = 3x - 10$

B $y = -3x + 23$

C $y = -3x + 38$ ✓

D $y = \frac{1}{3}x + 11$

WORKED SOLUTION & VERIFICATION

Midpoint $M = (9, 11)$ and gradient $m_{AB} = \frac{2}{6} = \frac{1}{3}$. The perpendicular gradient is $m_{\perp} = -3$, giving $y - 11 = -3(x - 9)$, which simplifies to $y = -3x + 38$.

20 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(8, 0)$ and $B(16, 4)$ (Segment #12).

A $y = \frac{1}{2}x + 2$

B $y = -2x + 26$ ✓

C $y = 2x - 13$

D $y = -2x + 17$

WORKED SOLUTION & VERIFICATION

Midpoint $M = (12, 2)$ and gradient $m_{AB} = \frac{4}{8} = \frac{1}{2}$. The perpendicular gradient is $m_{\perp} = -2$, giving $y - 2 = -2(x - 12)$, which simplifies to $y = -2x + 26$.

21 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(10, 2)$ and $B(12, 8)$ (Segment #13).

A $y = -\frac{1}{3}x + 19$

B $y = 3x + 5$

C $y = \frac{1}{3}x - 12$

D $y = -\frac{1}{3}x + \frac{26}{3}$ ✓

WORKED SOLUTION & VERIFICATION

Midpoint $M = (11, 5)$ and gradient $m_{AB} = \frac{6}{2} = 3$. The perpendicular gradient is $m_{\perp} = -\frac{1}{3}$, giving $y - 5 = -\frac{1}{3}(x - 11)$, which simplifies to $y = -\frac{1}{3}x + \frac{26}{3}$.

22 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(12, 4)$ and $B(16, 12)$ (Segment #14).

A $y = 2x + 8$

B $y = -\frac{1}{2}x + 25$

C $y = -\frac{1}{2}x + 15$ ✓

D $y = \frac{1}{2}x - 15$

WORKED SOLUTION & VERIFICATION

Midpoint $M = (14, 8)$ and gradient $m_{AB} = \frac{8}{4} = 2$. The perpendicular gradient is $m_{\perp} = -\frac{1}{2}$, giving $y - 8 = -\frac{1}{2}(x - 14)$, which simplifies to $y = -\frac{1}{2}x + 15$.

23 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(0, 6)$ and $B(6, 8)$ (Segment #15).

A $y = -3x + 13$

B $y = \frac{1}{3}x + 7$

C $y = 3x - 4$

D $y = -3x + 16$ ✓

WORKED SOLUTION & VERIFICATION

Midpoint $M = (3, 7)$ and gradient $m_{AB} = \frac{2}{6} = \frac{1}{3}$. The perpendicular gradient is $m_{\perp} = -3$, giving $y - 7 = -3(x - 3)$, which simplifies to $y = -3x + 16$.

24 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(2, 8)$ and $B(10, 12)$ (Segment #16).

A $y = -2x + 19$

B $y = \frac{1}{2}x + 10$

C $y = 2x - 7$

D $y = -2x + 22$ ✓

WORKED SOLUTION & VERIFICATION

Midpoint $M = (6, 10)$ and gradient $m_{AB} = \frac{4}{8} = \frac{1}{2}$. The perpendicular gradient is $m_{\perp} = -2$, giving $y - 10 = -2(x - 6)$, which simplifies to $y = -2x + 22$.

25 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(4, 10)$ and $B(6, 16)$ (Segment #17).

A $y = -\frac{1}{3}x + 21$

B $y = -\frac{1}{3}x + \frac{44}{3}$ ✓

C $y = 3x + 13$

D $y = \frac{1}{3}x - 6$

WORKED SOLUTION & VERIFICATION

Midpoint $M = (5, 13)$ and gradient $m_{AB} = \frac{6}{2} = 3$. The perpendicular gradient is $m_{\perp} = -\frac{1}{3}$, giving $y - 13 = -\frac{1}{3}(x - 5)$, which simplifies to $y = -\frac{1}{3}x + \frac{44}{3}$.